Identifiability of co-transcriptional splicing rates from steady-state isoform distributions

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Abstract

We study when kinetic rates of a co-transcriptional splicing system can be recovered from a single steady-state observation of its isoform proportion distribution (IPD). The system is modelled as a continuous-time Markov chain on a directed acyclic graph (the DADA graph) whose nodes are partial-splicing intermediates and whose edges fire with unknown rates. We parametrise families of rate vectors by a low-dimensional splicing regime vector and ask, for each region of regime space, whether the inverse map from IPD to rates is well-posed. Recovery from a single IPD turns out to be possible only in trivial cases (one intron) or under restrictive structural assumptions (Kinetic regime, where all rates within a junction are equal). For richer regimes, two distinct levels of structural information restore identifiability: knowing the active subgraph, or observing the same gene under multiple export contexts.

1 Introduction

[Motivate co-transcriptional kinetic competition and why rates *matter beyond outcome frequencies. Contrast with sequence-based alternative-splicing predictors. State the gap that motivates the inverse-problem framing.]*

The aim of this note is conceptual, not algorithmic. We do not propose a new estimator. We characterise the well-posedness of an inverse problem that several existing methods (NNLS-based and otherwise) tacitly assume to be solvable, and we work out what additional information — structural priors, multiple observations of the same gene — is needed to make the problem well-posed in cases where it is not.

2 The DADA graph

2.1 Splice sites, junctions, and compatibility

A pre-mRNA carries an alternating sequence of donor (D) and acceptor (A) splice sites. We write the gene's site sequence as a string in the alphabet $\{\mathsf{D}, \mathsf{A}\}$ — the *DADA string*. A *junction* is an ordered pair $j = (d, a)$ of a donor at site d and a later acceptor at site $a > d$; junction j corresponds to the molecular event of splicing out the intron between sites d and a.

Two junctions $j = (d, a)$ and $j' = (d', a')$ are *compatible* if (a) they share no splice site — $d \neq d'$ and $a \neq a'$ — and (b) their open intervals (d, a) and (d', a') are either disjoint or properly nested.

2.2 States, edges, and isoforms

Define the state set $\mathcal{V}$ as the collection of *antichains* of compatible junctions: each $S \in \mathcal{V}$ is a set of mutually compatible junctions. The biological reading is that S records exactly the splices that have already been made; the molecule in state S is the partial isoform with those

introns removed. The state $\emptyset$ is the unspliced pre-mRNA; the maximal antichains are mature isoforms.

The DADA graph is

$$G \;=\; (\mathcal{V}, \mathcal{E}), \qquad \mathcal{E} \;=\; \big\{\, (S, S \cup \{j\}) \;:\; S \in \mathcal{V},\; j \notin S,\; j \text{ compatible with all of } S \,\big\}.$$

Each edge fires a single junction. Edges are written $e = S \to S \cup \{j\}$, and we say edge e is *labelled by* junction $j(e) = j$; multiple edges may share a junction label, since the same splicing reaction can fire from different intermediate isoforms. The rate of edge e is $r_e \geq 0$. The graph is rooted at $\emptyset$ and is, by construction, a directed acyclic graph (DAG): each edge strictly enlarges the set of completed splices.

2.3 Worked example: `DADA` ($K = 2$ introns)

For the DADA string `DADA` (sites $\mathsf{D}_0\,\mathsf{A}_1\,\mathsf{D}_2\,\mathsf{A}_3$) the junctions are

$$j_1 = (0,1), \qquad j_2 = (0,3), \qquad j_3 = (2,3).$$

Compatibility:

- j_1 and j_2 share donor D_0: incompatible.
- j_2 and j_3 share acceptor A_3: incompatible.
- $j_1 = (0,1)$ and $j_3 = (2,3)$: disjoint intervals: compatible.

The state set $\mathcal{V} = \{\, \emptyset,\; \{j_1\},\; \{j_2\},\; \{j_3\},\; \{j_1, j_3\} \,\}$ has $N = 5$ elements; the graph has $M = 5$ edges:

$$\begin{aligned} e_1 : \emptyset \to \{j_1\},\; j(e_1) = j_1, \quad e_2 : \emptyset \to \{j_2\},\; j(e_2) = j_2, \\ e_3 : \emptyset \to \{j_3\},\; j(e_3) = j_3, \quad e_4 : \{j_1\} \to \{j_1, j_3\},\; j(e_4) = j_3, \\ e_5 : \{j_3\} \to \{j_1, j_3\},\; j(e_5) = j_1. \end{aligned}$$

Note that junctions j_1 and j_3 each label two edges (different intermediate states from which the same splicing reaction can fire), while j_2 labels only one edge because it is incompatible with everything else.

[TODO: Insert figure: DADA state graph with nodes $\emptyset, \{j_1\}, \{j_2\}, \{j_3\}, \{j_1, j_3\}$, edges labelled by $j(e)$.]

3 The forward model

Each non-root state i is exported (transcribed off the chromatin and observed as mature mRNA) at a known rate $\lambda_i \geq 0$. Pre-mRNA enters the system at the root $i = 0$ at unit production rate. The probability p_i of finding a single molecule in state i at steady state satisfies, for each $i \in \mathcal{V}$,

$$\underbrace{\sum_{e:\ \mathrm{dst}(e)=i} r_e\, p_{\mathrm{src}(e)}}_{\text{flux in}} \;-\; \underbrace{\sum_{e:\ \mathrm{src}(e)=i} r_e\, p_i}_{\text{flux out}} \;-\; \lambda_i\, p_i \;=\; -\delta_{i,0}. \tag{1}$$

The root row is dependent on the others (mass conservation: total inflow = total export), so we drop it and stack the remaining $N-1$ equations into matrix form,

$$A(p)\, r \;=\; c(p), \qquad A \in \mathbb{R}^{(N-1)\times M}, \qquad c \in \mathbb{R}^{N-1}, \tag{2}$$

with

$$A_{i,e} \;=\; \begin{cases} +\,p_{\mathrm{src}(e)} & \text{if } \mathrm{dst}(e) = i, \\ -\,p_{\mathrm{src}(e)} & \text{if } \mathrm{src}(e) = i, \\ 0 & \text{otherwise,} \end{cases} \qquad c_i \;=\; \lambda_i\, p_i.$$

Equation (2) is the central object. We observe p (the IPD), λ is known, so A and c are computable; the rates r are the unknowns.

4 The inverse problem

Given the IPD p and the export rates λ, we want to recover the rate vector r. A natural formulation is

$$\widehat{r} \;=\; \arg\min_{r\geq 0} \; \big\|A(p)\,r - c(p)\big\|_2^2. \tag{3}$$

The non-negativity constraint is biologically essential: rates are physical reaction frequencies. We refer to the optimisation problem (3) as the *inverse problem*; the question of this paper is whether $\widehat{r}$ equals the truth $r^\star$, and under what conditions.

5 The splicing regime vector

5.1 Generative model for rates

We model the truth $r^\star$ as a draw from a parametric family controlled by a low-dimensional regime vector

$$\theta \;=\; (\eta,\ \sigma,\ \alpha) \;\in\; [0,1]\times\mathbb{R}_{\geq 0}\times\mathbb{R}_{>0}.$$

The Mixed-Dirichlet generator at θ samples a rate vector by:

1. For each junction j, draw an amplitude $\rho_j \sim \mathrm{LogNormal}(\log R_0,\ \sigma^2)$.
2. For each non-root state i with m_i incoming edges, set the active cardinality

   $$k_i \;=\; \max\big(1,\ \mathrm{round}(\eta\, m_i)\big),$$

   pick a uniform random size-k_i subset S_i of the m_i incoming edges, and draw weights $w \sim \mathrm{Dirichlet}(\alpha,\ldots,\alpha)$ over S_i.
3. Set $r_e = \rho_{j(e)}\cdot k_i \cdot w_e$ if $e\in S_i$ (where $i=\mathrm{dst}(e)$), else $r_e = 0$.

The active mask is $S^\star = \{e : r_e > 0\}$ and the active subgraph $G^\star$ is the subgraph induced on edges in $S^\star$.

5.2 Four canonical regimes

The corners and faces of regime space have biologically natural interpretations:

- **Null**: $\sigma = 0$, $\alpha\to\infty$. Degenerate: all rates equal R_0. Useful as a baseline.
- **Predetermined**: $\eta = 0$. Each child has exactly one active parent, so $G^\star$ is a tree (a spanning tree of the active state subgraph).
- **Kinetic**: $\eta = 1$, $\alpha\to\infty$. Every edge is active and the Dirichlet weights are uniform within each junction, so $r_e = \rho_{j(e)}$ for every active edge of junction j.
- **Mixed**: $\eta\in(0,1)$, α moderate. Partial branching with uneven weights. The biologically realistic regime, and the hard one for inverse inference.

The four are not exhaustive; they are landmarks. The whole prior cube $\theta\in[0,1]\times[0,\infty)\times(0,\infty)$ is a continuous interpolation between them.

6 What is identifiable, and when

We work throughout this section with a single noiseless IPD p and known export rates λ.

6.1 The relevant linear-algebraic objects

Given an active edge set $\mathcal{E}^\star$ define the active subgraph $G^\star = (\mathcal{V}^\star, \mathcal{E}^\star)$ where $\mathcal{V}^\star$ is the set of states touched by edges in $\mathcal{E}^\star$ together with the root; let $N^\star = |\mathcal{V}^\star|$ and $M^\star = |\mathcal{E}^\star|$. We assume throughout that $G^\star$ is connected (always true for our DADA graphs because of the rooted construction).

The masked balance matrix $A_\star$ is the column restriction of A in (2) to columns indexed by $\mathcal{E}^\star$. It admits the factorisation

$$A_\star \;=\; B_\star \cdot D_\star,$$

where $B_\star$ is the (root-removed) signed incidence matrix of $G^\star$ — with $(B_\star)_{i,e} = +1$ if $\mathrm{dst}(e) = i$, -1 if $\mathrm{src}(e) = i$, and 0 otherwise — and $D_\star$ is the diagonal matrix with $(D_\star)_{e,e} = p_{\mathrm{src}(e)}$. When all $p_{\mathrm{src}(e)} > 0$, $D_\star$ is invertible and

$$\mathrm{rank}(A_\star) \;=\; \mathrm{rank}(B_\star) \;=\; N^\star - 1.$$

This last equality is the standard fact that the (reduced) signed incidence matrix of a connected graph on $N^\star$ vertices has rank $N^\star - 1$.

The kernel of $A_\star$ is therefore

$$\ker(A_\star) \;=\; D_\star^{-1}\ker(B_\star) \;\cong\; \ker(B_\star),$$

which is the cycle space of $G^\star$ (treated as an undirected graph for cycle-space purposes), of dimension equal to the cyclomatic number

$$c(G^\star) \;=\; M^\star - N^\star + 1. \tag{4}$$

A non-trivial element of $\ker(B_\star)$ corresponds to a cycle in the underlying undirected graph of $G^\star$ — equivalently, two distinct directed paths between the same pair of states. Although $G^\star$ has no directed cycles (it is a DAG), the existence of multiple parallel directed paths between the same pair of vertices means rate flux can be *redistributed* between alternatives without changing the IPD.

6.2 Single IPD with known active subgraph

If we are given the exact active set $\mathcal{E}^\star$, the inverse problem reduces to solving (3) restricted to columns of $\mathcal{E}^\star$. The picture splits cleanly on $c(G^\star)$:

Tree case ($c(G^\star) = 0$). Then $M^\star = N^\star - 1 = \mathrm{rank}(A_\star)$, so $A_\star$ has full column rank, the system $A_\star r = c$ has at most one solution, the truth $r^\star$ is one such, and NNLS returns $r^\star$ exactly.

Cyclic case ($c(G^\star) > 0$). Then $\ker(A_\star)$ has positive dimension. For a generic truth $r^\star$ in the strictly positive orthant, $r^\star$ is in the relative interior of the polytope

$$\{\, r \geq 0 \;:\; A_\star r = c \,\},$$

which has dimension $c(G^\star) > 0$. Multiple non-negative rate vectors then produce the same IPD, and NNLS alone (without further structure) cannot distinguish them. The corner of the polytope picked by Lawson–Hanson NNLS depends on algorithmic details rather than on the data.

Numerical aside. Even in the tree case, plain NNLS can be numerically poor on graphs of large depth because $A_\star = B_\star D_\star$ inherits a large condition number when source-state probabilities span many orders of magnitude. We have observed that on the tree subgraph of `DADADADADADA` ($K = 6$ introns), per-edge NNLS frequently fails despite full column rank. The augmented system used by junction-shrinkage NNLS (Section 6.4) regularises the conditioning sufficiently to recover the truth at machine precision.

6.3 Single IPD without the mask

In the realistic case where we observe only p and the topology G (not which edges are active), the situation is strictly worse. The full balance matrix A has M columns and $N-1$ rows with $M > N-1$ for every non-trivial DADA graph; therefore

$$\dim\ker(A) \;\geq\; M-(N-1) \;>\; 0,$$

and the NNLS solution set has positive dimension regardless of $c(G^\star)$. The truth $r^\star$ is one of many non-negative rate vectors fitting the same p.

A separate issue is that distinct active subgraphs can produce identical IPDs. The *cardinality shell* of a realisation is the set of active masks consistent with the cardinality profile $(k_i)_{i\in\mathcal{V}^\star}$. We have observed empirically (*[on* `DADADA`*, $K=3$, with $\eta=0.5$, all 96 elements of the shell admit non-negative rate vectors fitting the same IPD to floating-point precision]*). Cardinality is identifiable from the IPD; mask identity, generically, is not.

6.4 Junction-shrinkage and the Kinetic regime

Suppose we additionally constrain the rates to lie on the *junction-equal subspace*

$$W_{\text{eq}} \;=\; \left\{\, r\in\mathbb{R}^{M^\star} \;:\; r_e = r_{e'} \text{ whenever } j(e)=j(e') \,\right\}.$$

Inside W_{eq}, the unknowns collapse from $M^\star$ edge rates to $J^\star$ junction rates, where $J^\star$ is the number of junctions touched by edges in $\mathcal{E}^\star$. The pooled balance matrix A_{pool} then has shape $(N-1)\times J^\star$, and for typical DADA topologies $J^\star \ll N-1$, so A_{pool} is generically full column rank.

Two consequences:

1. Pooled NNLS (which solves (3) on A_{pool} rather than A) recovers the junction rates uniquely from a single IPD. *No active mask is required*: an inactive edge contributes zero to the column-aggregation in its junction. This is the Kinetic regime ($\eta=1$, $\alpha\to\infty$), because the truth $r^\star$ then satisfies $r^\star\in W_{\text{eq}}$ exactly — all active edges of a junction share the rate ρ_j.

2. The junction-shrinkage NNLS solver augments the balance equations with Helmert contrasts, equivalent to a soft penalty for deviation from W_{eq}. The augmented matrix is overdetermined regardless of $c(G^\star)$. When $r^\star\in W_{\text{eq}}$, the augmented system is consistent and recovers $r^\star$ exactly (this includes both the Predetermined regime, where every junction has at most one active edge so the constraints are vacuous, and the Kinetic regime). When $r^\star\notin W_{\text{eq}}$, the augmented system is inconsistent and the LS solution is biased toward W_{eq}, giving $\widehat{r}\neq r^\star$.

The boundary between "junction-shrinkage works" and "junction-shrinkage fails" is therefore not the cyclomatic boundary $c(G^\star)=0$, but the inclusion $r^\star\in W_{\text{eq}}$. Under the Mixed-Dirichlet generator, $\Pr(r^\star\in W_{\text{eq}})\to 1$ as $\alpha\to\infty$ (the Kinetic limit) and $\Pr(r^\star\in W_{\text{eq}})=1$ when $\eta=0$ (Predetermined, vacuous constraints). For Mixed regimes at finite α, the probability is generically zero.

6.5 Worked example: $K=2$ (`DADA`)

For `DADA` (Section 2.3) the full graph has $M=5$, $N=5$, and

$$c(G) \;=\; M-N+1 \;=\; 1.$$

The single cycle is the diamond at state $\{j_1,j_3\}$, reachable through $\{j_1\}\xrightarrow{e_4}\{j_1,j_3\}$ or through $\{j_3\}\xrightarrow{e_5}\{j_1,j_3\}$. Concrete consequences:

- **Predetermined, or Mixed at $\eta < 0.75$.** The branching state $\{j_1, j_3\}$ has $m = 2$ parents, and the active cardinality $k = \max(1, \mathrm{round}(2\eta))$ equals 1 for $\eta < 0.75$. Only one of $\{e_4, e_5\}$ is active. The active subgraph is a 4-edge tree ($c(G^\star) = 0$). Identifiable *given the mask*, but not from the IPD alone — the cardinality shell of size 2 (active edge e_4 *or* e_5) is the source of ambiguity.
- **Kinetic ($\eta = 1$, $\alpha \to \infty$).** All 5 edges active, $r^\star \in W_{\mathrm{eq}}$, pooled NNLS solves a 4×3 system with unique solution. Identifiable from the IPD alone, no mask required.

The trivial graph `DA` ($K = 1$, $N = 2$, $M = 1$, $c = 0$) is identifiable under any prior because the single rate is read off from the single non-root balance equation directly.

7 Paths to enlarge identifiability

Three structural augmentations restore identifiability outside the cases of Section 6. We describe each briefly; each suggests a different experimental or computational direction.

7.1 Multiple export contexts

[Notation: we use K for the number of introns elsewhere; here we use T (number of tissues / contexts) to avoid the clash.]

Suppose we observe the same gene's IPD under T distinct export profiles $\lambda^{(1)}, \ldots, \lambda^{(T)}$ — e.g. the same gene measured in different cell types or conditions, in which the underlying junction rates r are conserved but per-state decay rates differ. Stacking the T balance systems yields

$$\underbrace{\begin{bmatrix} A^{(1)}(p^{(1)}) \\ \vdots \\ A^{(T)}(p^{(T)}) \end{bmatrix}}_{T(N-1) \times M^\star} r \;=\; \begin{bmatrix} c^{(1)} \\ \vdots \\ c^{(T)} \end{bmatrix}.$$

Each block alone has rank $N^\star - 1$, but the column scalings $D_\star^{(t)}$ differ across blocks because the $p^{(t)}$ differ; the kernels $\ker(A_\star^{(t)})$ are accordingly different "rotations" of the same underlying cycle space $\ker(B_\star)$. Generically, $T \geq \lceil M^\star/(N^\star - 1) \rceil$ contexts suffice to make the stacked matrix full column rank, recovering $r^\star$ exactly without knowing the active mask. For most practical DADA topologies this number is small (2 or 3).

[We should validate this dimension bound on real data — to what extent is GTEx- or ENCODE-level cell-type variation generic *in the sense required?]*

7.2 Sequence-based priors on active junctions

Pre-trained sequence models (SpliceAI, AlphaGenome) predict per-junction probabilities of splice-site usage from sequence alone. Thresholding these declares some junctions inactive, eliminating their edges from $\mathcal{E}$ and shrinking $M^\star$. A prior strong enough to bring $M^\star$ down to $N^\star - 1$ recovers the tree case directly. Even partial reductions help: each removed edge can decrease $c(G^\star)$ by 0 or 1, and the cardinality shell shrinks correspondingly.

7.3 Direct path observation (long-read)

Long-read sequencing observes individual transcript paths through the DADA graph rather than aggregate isoform proportions. Empirical edge frequencies from many reads give a direct estimate of the active mask $S^\star$. Combined with NNLS on the masked balance system, this is the cleanest route in current data modalities and arguably the most under-used.

8 Summary

The state of identifiability is summarised in Table 1.

	Single IPD		**Multi-context IPDs**	
Splice regime	without mask	with mask	without mask	with mask
Trivial ($K = 1$, `DA`)	yes	yes	yes	yes
Predetermined / Mixed ($K \geq 2$)	no	yes (tree)	yes *[check]*	yes
Kinetic	yes (junction-equal)	yes	yes	yes

Table 1: Identifiability of the rate vector $r^\star$ as a function of the splice regime and the structural information available. "Tree" means the active subgraph $G^\star$ has cyclomatic number zero; "junction-equal" means the truth lies in W_{eq}. The mark "*[check]*" indicates an empirical claim we have not yet rigorously verified.

The single most important takeaway: *from a single IPD with no structural prior, the only splice regime that admits identifiable rates is Kinetic*, because that is the only regime where the truth's structure (junction-equal) collapses the $M^\star$-dimensional inverse problem to the lower-dimensional pooled problem on $J^\star$ junction rates. Every other regime requires either the active mask, or multiple observations from different export contexts, or a sequence-based prior that shrinks the candidate active edge set.

9 Open questions

- How non-generic are real cell-type-paired export profiles? Does GTEx- or ENCODE-level across-tissue variation suffice to satisfy the multi-context recovery condition for typical genes?
- How sharp is the "junction-equal" boundary for biological data? In real splicing, within-junction weight ratios are presumably non-uniform; how close to equal is "equal enough" for junction-shrinkage NNLS to recover useful (if biased) rates?
- What is the right way to combine sequence-based junction priors with the multi-context approach to handle the largest practical DADA graphs ($K \geq 8$ introns), where exhaustive enumeration of cardinality shells or active subgraphs is infeasible?
- Are there genes whose splicing structure reliably falls in the Kinetic regime in some tissues? For those, single-IPD inversion via pooled NNLS is exact, with no auxiliary information needed — a directly testable prediction.

References

[1] Martin Slawski and Matthias Hein. Non-negative least squares for high-dimensional linear models: Consistency and sparse recovery without regularisation. *Electronic Journal of Statistics*, 7:3004–3056, 2013.

[2] *[nano-COP / Churchman lab paper — fill in correct reference.]*

[3] *[Drexler et al. 2020 on K562 chromatin nano-COP — fill in.]*

[4] *[Avsec et al., AlphaGenome — fill in.]*

[5] Norman Biggs. *Algebraic Graph Theory.* Cambridge University Press, 2nd edition, 1993. (Standard reference for cycle space, signed incidence matrix rank, and the cyclomatic number.)